Mike Wolak · mike@epromfoundry.com · (512) 944-8586

Paula and Toccata Audio of the G2 Amiga

Paula's four channels and a Toccata 16-bit card mixed in the fabric at 48 kHz and played through the G2's WM8750BL codec; an AHI driver gives AmigaOS programs the Toccata's 16-bit modes.

HIGHLIGHTS

- Paula and Toccata mixed at 48 kHz.
- AHI 6 driver g2.audio: four modes, 16 rates up to 48 kHz.
- 18-bit mix FIFO of 1,024 entries, interrupt-driven refill of the codec.

KEY PARAMETERS

PARAMETER	VALUE
Mix rate	48 kHz
Codec	WM8750BL, 24-bit I²S
Mix FIFO	1,024 entries
Toccata rates	16, to 48 kHz
AHI modes	4

1 AUDIO PATH

STAGE	IMPLEMENTATION
Sources	Paula's four channels (two 15-bit outputs, doubled to 16 bits) and the Toccata card (AD1848 model, 16-bit stereo, 16 rates)
Mix	An NCO in core time takes a stereo sample at 48 kHz into a 1,024-entry FIFO (18-bit samples, read saturated to 16 bits)
Firmware	An interrupt at the FIFO's watermark (256) drains it, resamples to the codec's exact rate and refills the codec FIFO
Codec	Wolfson WM8750BL, I²S 24-bit, the codec the slave, 48 kHz; the G2's audio output jack
Rate	NTSC: AUD_INC 0x006D_D9CF (48 kHz in 28.636364 MHz core time); PAL: 0x006E_DCAE

2 TOCCATA AND AHI

ITEM	VALUE
Board	Toccata Zorro II (manufacturer 18260, product 12), register access as the MacroSystem card; playback only
Driver	DEVS:AHI/g2.audio with the mode file DEVS:AudioModes/G2 (AHI 6.x)
Modes	0x00470001 16 bit stereo++, 0x00470002 HiFi 16 bit stereo++, 0x00470003 16 bit mono, 0x00470004 HiFi 16 bit mono
Rates	The 16 AD1848 rates (5,513 to 48,000 Hz), from a 32-bit phase accumulator in core time, exact in the mean
Buffering	At 44.1 kHz with a 50 Hz player: 882 frames a pass, card FIFO 1,024 bytes (5.8 ms)
Measured	AHI Preferences lists the modes and plays its test tone; Play16 1.10 plays a 44.1 kHz 16-bit WAV to the end

3 LIMITS AND UNTESTED ITEMS

ITEM	STATE
Recording (Toccata input)	Not supported
Audio quality figures (THD, noise)	Not measured
Toccata through toccata.library programs	Untested: the driver programs the card directly

Measured on a G2 revision C unit with firmware release 1.0 and fabric 0x0AB5EF2D unless stated. Repository mikewolak/pano-g2-amiga, release 1.0.